Spotify®

AI-Powered Solution to the Freemium Paradox

Viet Career Guide Think-A-Thon Case Competition 2026

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Executive Summary

Increase Premium conversion by delivering the right offer to the right user at the right moment

What is the goal?

Increase Free-to-Premium conversion over the next 12 months using AI, while protecting the Free ecosystem and long-term user growth.

What is the problem?

Only 38% of Spotify's 761M MAUs convert to Premium. Existing systems decide *whether* to message, never *what* to offer or *when*, and treat all free users identically, even though **47% will never convert** and **100% of converters hit a friction wall first**. Spotify's own Audiobooks+ proves this exact signal already creates value, found once, by accident, in a single vertical.

How do we solve this problem?

The Uplift Engine

System 1
Smart Conversion Engine
Lift conversion

System 2
Margin Mix Engine
Improve profitability

System 3
Growth & Retention Engine
Sustain long-term growth

How do we implement?

Build and validate System 1 within four quarters. Then **extend the same platform** to Systems 2 and 3 through a staged, four-gate rollout, scaling each only after it proves incremental economics.

What do we gain?

850M MAUs in 2027E.
By 2030E: **1.001B MAUs, 35.3% gross margin, 20.1% operating margin, €253M NPV, 290% ROI**

Industry

Streaming growth is accelerating, not just continuing: 6.8% CAGR the past four years, projected to reach 11.1% through 2030

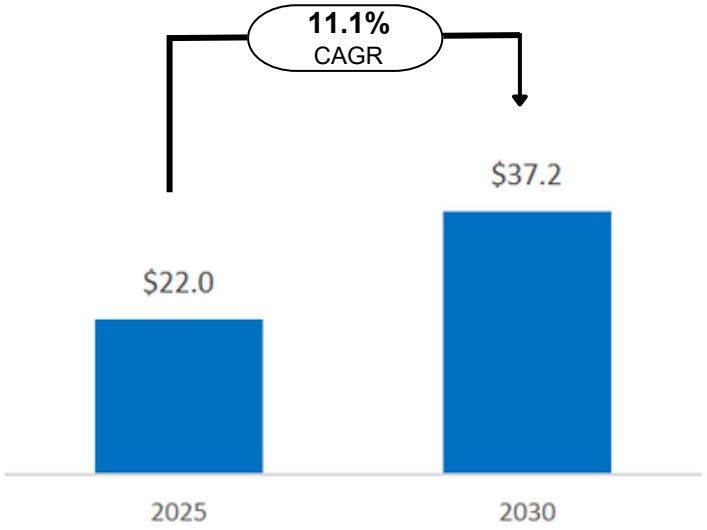
Market grew at a 6.8% CAGR from 2021 to 2025, reaching $22.0B

Global Music Streaming Market Value from 2021 to 2025, *in $ billion*



Market is projected to grow at an 11.1% CAGR from 2025 to 2030, reaching $37.2B

Global Music Streaming Market Value 2030 Projection, *in $ billion*



Key Growth Drivers

- Growth of streaming services and digital media consumption
- Widespread adoption of smart devices
- **Personalization and recommendations**
- Expansion into podcasts, audiobooks, and exclusive content
- Technological innovations such as voice assistants and spatial audio

***Personalization and recommendations*, a growth driver through higher engagement and retention**

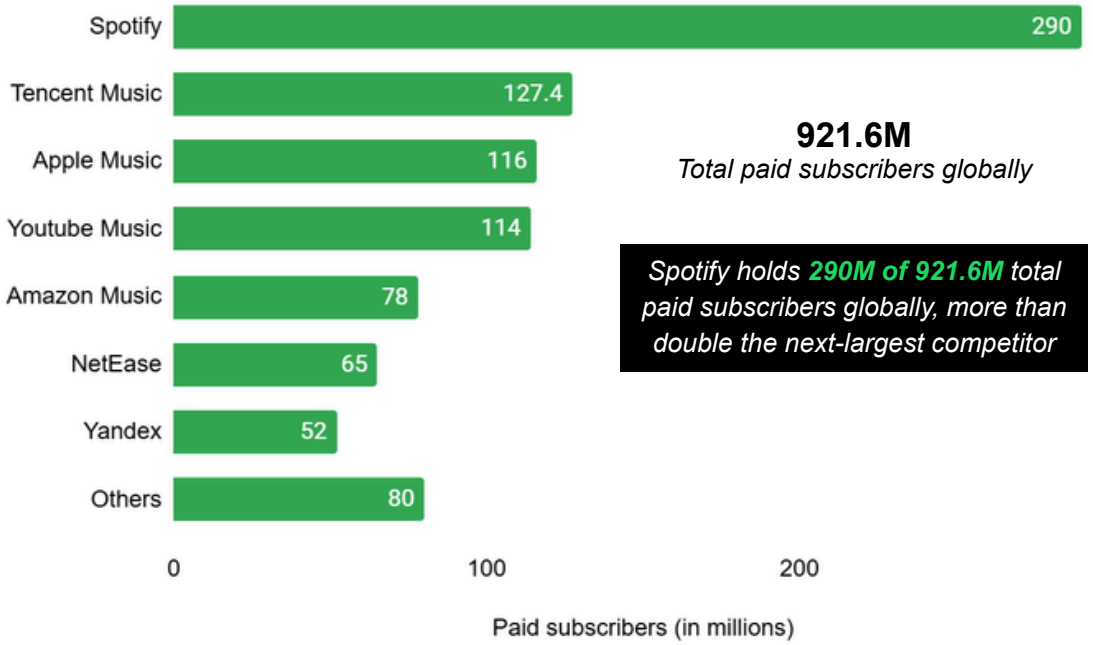
Source: IFPI, Goldman Sachs

Industry

Spotify holds a stable ~31% share for five straight years, defending its lead as the market grows.

Spotify captures nearly one-third of the global paid music streaming market.

Music streaming paid subscribers by Q4 2025, *in million paid subscribers*



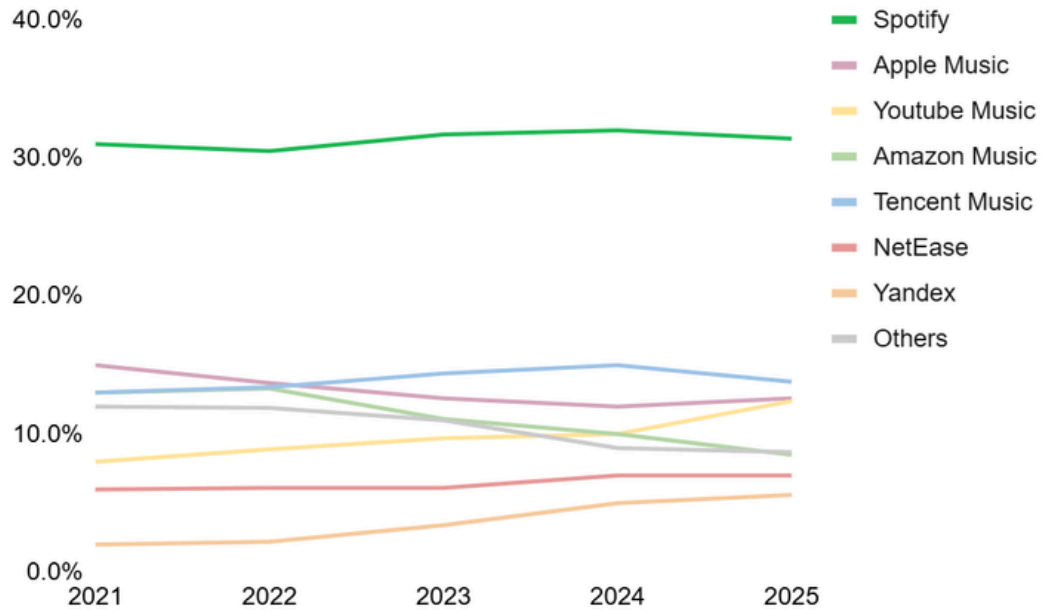
921.6M

Total paid subscribers globally

*Spotify holds **290M of 921.6M** total paid subscribers globally, more than double the next-largest competitor*

Spotify continues to lead the industry by a wide margin since 2021

Music Streaming Market Share by Platform from 2021–2025, *in percentage*



Spotify has held a stable share for five years, leading the market despite active competition from major competitors like Apple, Tencent, and YouTube Music.

Source: MIDiA Research

Industry

Spotify doesn't win on every dimension, it wins on personalization and discovery

Competitive Analysis

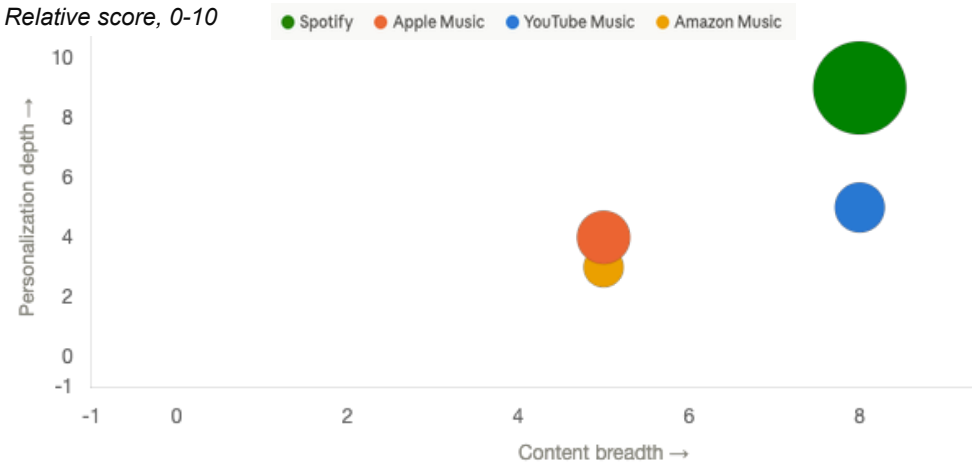
Platform	Price	Key Differentiator	Competitive Position	Market Share
Spotify	$11.99/month	AI recommendations (Discover Weekly, AI DJ), Spotify Connect, Wrapped	Leader in personalization & music discovery	31.4%
Apple Music	$10.99/month	Lossless & Spatial Audio, Apple Classical, Apple Sing	Best for Apple ecosystem users	12.6%
YouTube Music	$10.99/month	Largest music/video library, unofficial & live content	Strong YouTube ecosystem integration	12.4%
Amazon Music	$10.99/month	Alexa integration, HD/Ultra HD audio	Best value for Prime members	8.5%

Pricing is largely standardized across major platforms. Customer choice, therefore, depends more on **differentiated user experiences**.

Spotify doesn't win on every dimension. It bet on discovery and breadth over audio fidelity and hardware lock-in, and the share numbers show that bet paid off.

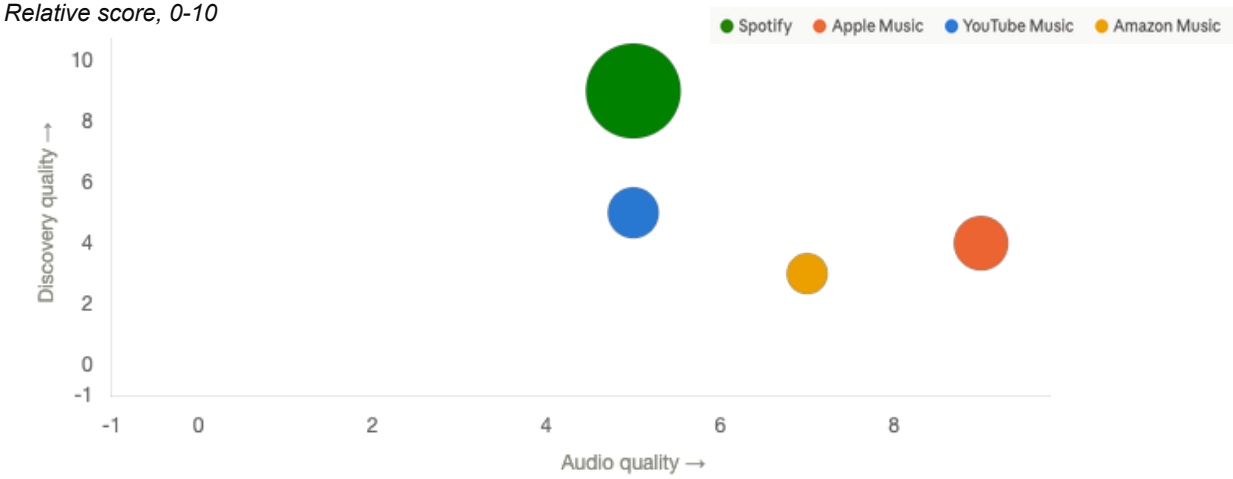
Spotify is strong on both content breadth and personalization

Relative score, 0-10



Spotify trades audio fidelity for discovery depth, and the market has rewarded that trade

Relative score, 0-10



Source: Spotify, Apple Music, Youtube Music, Amazon Music

Company Overview

Spotify is a global platform with a vast content catalog and reach across every major region

About the Company

Spotify, launched in 2008, is a **global audio streaming platform** offering on-demand music, podcasts, and audiobooks, distinguished by AI-driven personalization that tailors the listening experience to each user.

100M+
Music tracks

7M+
Podcasts

700K+
Audiobooks

Mission: To deliver creativity to the world–one note, one voice, one idea at a time.

Two Services: Ad-Supported & Premium

Spotify operates on a *freemium model*, offering two ways to access its platform: **Ad-supported tier** and a paid **Premium subscription**.

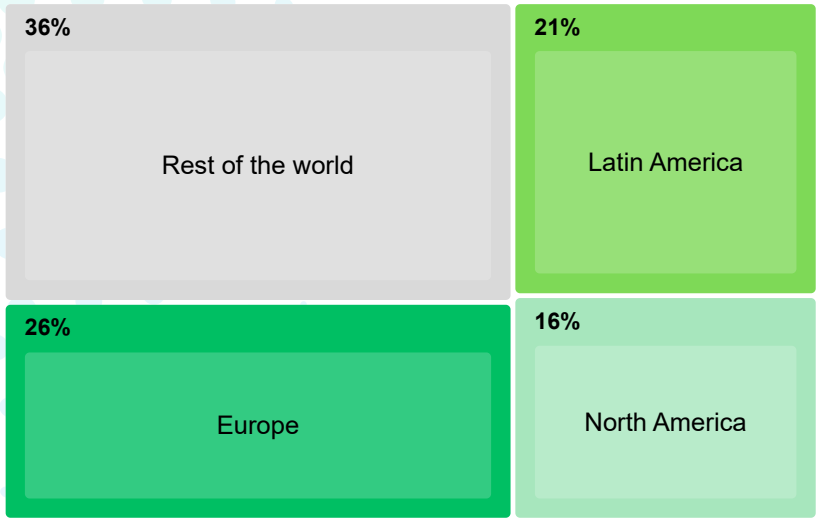
Ad-Supported

Free access *to a* ***full catalog of music, podcasts,*** *and* ***select audiobook content****, monetized through audio, video, and display ads.*

Premium Subscription

Paid subscription *that provides* ***ad-free*** *and* ***full control*** *over listening experience and* ***additional personalized features***

Major Global Markets



237
Countries and Territories

184
Markets where Premium is supported

Source: Spotify 20F, Spotify Website

Company Overview

Ad-supported and Premium has different value propositions, and are two fundamentally different ways Spotify gets paid

Ad-Supported & Premium Feature Comparison

Feature	Ad-supported	Premium
Ad-free	✗	✓
Skips	*Limited*	*Unlimited*
Offline downloads	✗	✓
Audio quality	*Standard*	*Higher/lossless*
Podcasts	*Ad-supported*	*Ad-free*
Audiobooks	*Limited/preview*	*Monthly hours included*

Other Special Premium Features

AI DJ: AI-narrated, personalized listening session

Prompted Playlist: AI builds a playlist matching users' descibed moods and vibes

SongDNA: Explore the writers, producers, samples, and interpolations behind a track

Taste Profile: Users can view how Spotify's algorithm interprets their listening habits and refine what shows up on their homepage

Ad-Supported Monetization Engine

Revenue is generated from ads

Advertiser → pays Spotify per ad impression (CPM) → Spotify targets and inserts the ad → Free listeners

*Spotify then pays **royalties** to rights holders for the track played*

Premium Monetization Engine

Revenue is generated from subscriptions

Subscribers → pays a flat monthly fee → Premium features, no ads

*Spotify then pays **royalties** to rights holders for the track played*

Subscription Plans – *U.S. market*

Individual	**Duo**	**Family**	**Student**
$12.99	*$18.99*	*$21.99*	*$6.99*
Single user	2 people	Up to 6	Verified student

**Premium pricing is regionally adjusted, not a flat global rate*

Both tiers pay the same royalty costs per stream, the difference is entirely on the revenue side. Ad-supported tier has a thin ad-based fee, while premium has a larger subscription fee.

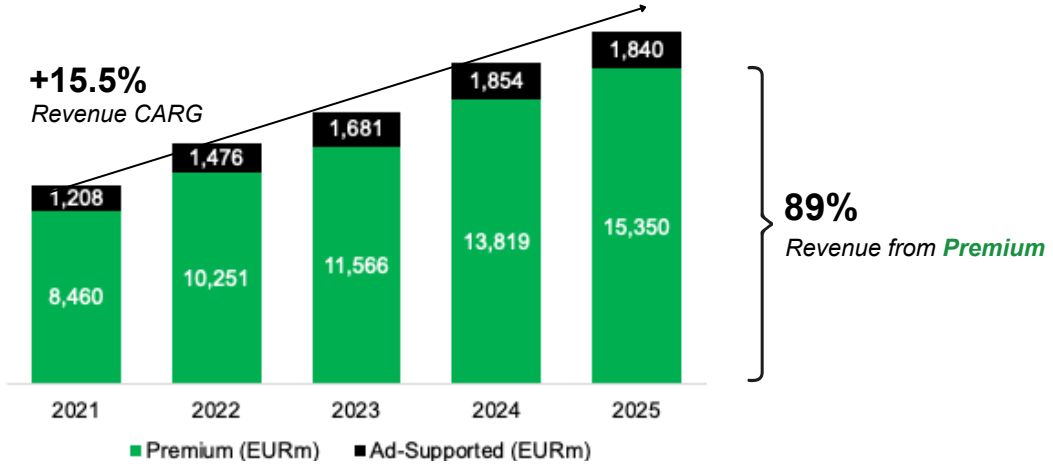
Source: Spotify 20F, Spotify Website

Company Overview

Ad-Supported acquires. Premium converts. That gap is the opportunity.

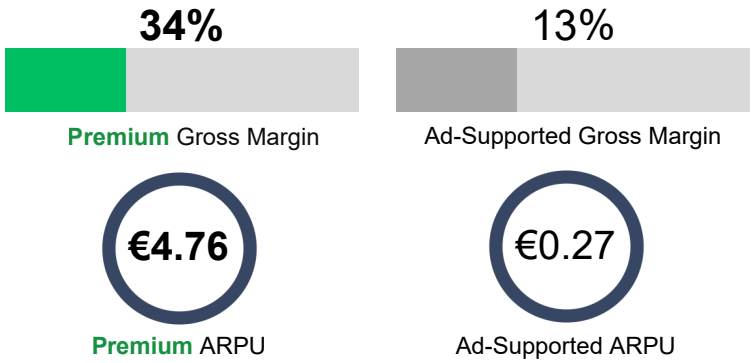
Premium drives 89% of revenue and growth is already tracking to 2030 target

Spotify's Revenue and Profit Margin Growth over 5 years, in EUR millions



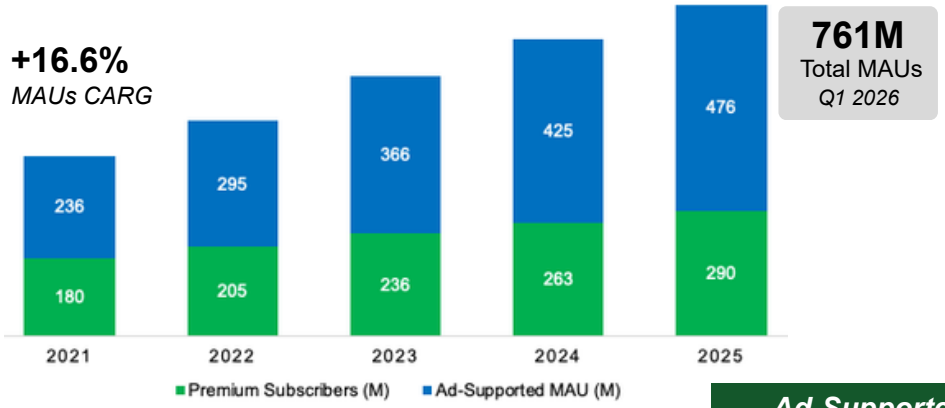
Premium is the profit engine (2.7x margin, 18x ARPU)

Premium vs. Ad-supported Gross margin and ARPU



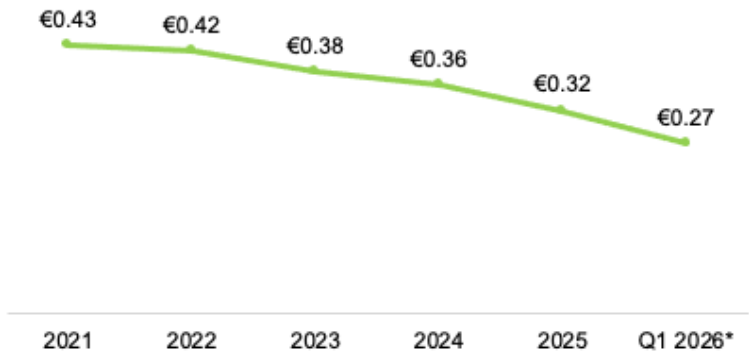
Ad-Supported users are growing faster than Premium...

Spotify's Monthly Active Users over 5 years, in millions



...but each free user is worth 37% less than in 2021

Ad-Supported ARPU (€/month)



Ad-Supported is the acquisition engine. Premium is where profit sits. Closing the gap between them, through conversion, is the opportunity.

Source: Spotify 20F, Spotify Newsroom

By 2030

1B
Monthly Active Users

Mid-teens
Revenue Growth

35-40%
Gross Margin

> 20%
Operating Margin

Spotify's *"North Star"* Goals

1B	*Subscriber*
100B	*Annual Revenue*
40%	*Gross Margin*

Problem Diagnosis

Existing systems optimize for engagement and decide whether to prompt, not what to offer or when it matters most.

Problem Diagnosis:

Why Conversion Isn't Working

Current Systems at Spotify:	The Gap:
BaRT – Spotify's contextual bandit system for Home screen recommendations. It decides which personalized shelves and cards to show, such as "Made for You," recent-listening shelves, and artist spotlights	The system is optimized for engagement, click-through, and consumption. It has **no conversion objective.**
Uplift messaging model *(Spotify Engineering, 2023)* **–** Spotify uses causal ML to decide whether to send a user a message, explicitly including Premium-prompting messages, based on predicted individual treatment effect.	The model decides whether to prompt, but **not what to offer**. Everyone who clears the gate sees the same "Go Premium" message and price, whether they're blocked by cost or by control.
Spotify's Announced AI: Large Taste Model, Remix, Prompted Playlists All of it is *generative*, and generative AI costs money per call.	The AI Spotify is missing is ***decisional***. Decisional AI does not generate content, but allocates offers. It is cheaper to run and it is directly margin-accretive.

*Neither systems answers the critical questions – **What** offer should this specific user see, and **when** exactly should it fire?*

Why "What" Matters

Users aren't blocked by the same thing.

Friction-frustrated
Need unlock

Price-sensitive
Care about pricing plans

Why "When" Matters

Friction is a moment, not a trait — it spikes, then fades.

Offer fires at the peak
User is primed, likely to accept

Offer fires next day
Frustration faded, offer lands flat

If the moment is missed, users find a workaround, tolerate, or move on.

Source: Spotify Engineering 2023

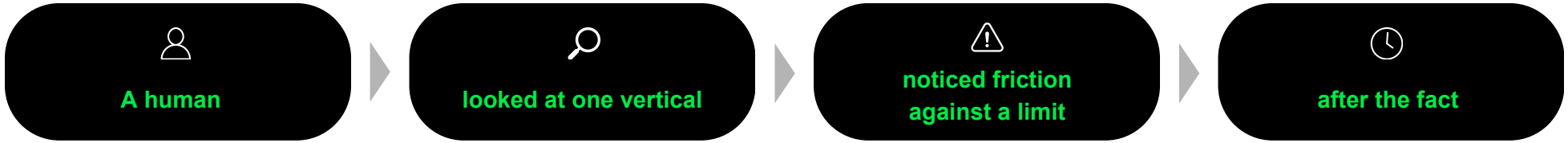
Problem Diagnosis

Audiobooks+ already proved this – Spotify just doesn't know it's a system

The Story Behind Audiobook+ Discovery – Investor Day 2026

At Investor Day 2026, Spotify Co-CEO Gustav Söderström recounted a story from earlier: when asked which add-on would appeal enough to make super-users pay more, he expected music. It turned out to be audiobooks, which was discovered when the team noticed a **segment of audiobook streamers hitting their monthly usage limits** more than other users, and built Audiobooks+ in response.

Spotify's most successful new add-on, Audiobook+, was discovered by:



Our survey found the identical mechanism at the conversion boundary



100%
of converters hit a limit the week before they upgraded



32%
of converters listened more

We asked current Premium users "Which behaviors were they doing more often the week before they upgraded"

100% – **All responders** reported at least one of these frictions:

- Searching against on-demand limits
- Hitting skip caps
- Wanting offline saves
- Exceeding ad tolerance

Primary survey, n=74, convenience sample skewed toward Premium and 18–24. Used to evidence behavioural mechanisms, not to estimate population rates.

Friction against a limit is the highest-value monetization signal Spotify owns, found by hand, once, in one vertical.

What if it were found automatically, everywhere, all the time?

Source: Spotify Investor Day 2026, Primary survey

Problem Diagnosis

Our primary survey evidence reveals at least three distinct behavioral patterns coexisting in the same free base

Within the exact same free-user population, there are different reasons someone isn't converting:

47% of free users said nothing would move them	→	**Group 1**: A group no offer will ever reach
100% of upgraders hit a friction wall first	→	**Group 2**: friction, not tenure, triggers conversion
1 in 3 cite price, not friction, as a blocker	→	**Group 3**: a separate reason entirely – conversion is price sensitive
3 in 6 churned right after a free trial or promo ended	→	**Group 4**: even converts don't behave alike

Key Insight:

Users respond differently, even within the same population. One prompt can't fit a group that won't respond, a group blocked by price, and a group blocked by friction.

→ Sent to the wrong group, it can backfire.

Primary survey, n=74, convenience sample skewed toward Premium and 18–24. Used to evidence behavioural mechanisms, not to estimate population rates.

Source: Spotify Investor Day 2026, Primary survey

Solution

The Uplift Engine

A two-stage AI system decides who to target, then how to convert them

Stage 1 finds who an offer can actually move, Stage 2 decides the rung, timing, and offer itself

Two-Stage AI Overview:

All Free Users (468M) → *STAGE 1* **Uplift segmentation finds the Persuadables** → **Persuadables** *Target Group: Free users that would only subscribed if nudged to* → *STAGE 2* **Escalation ladder decides rung, timing, and offer** → **Converts to Premium**

Stage 1: Identify "Who" – The "Persuadables"

Scores every free user on **whether an offer would actually change their behavior**, not just whether they're likely to convert, using the same causal, uplift-based approach Spotify already runs in production for in-app messaging *(Spotify Engineering, 2023)*.

Stage 2 – How & When

For each Persuadable, the model **checks one step at a time**, first naming their friction, then a feature preview, then a personalized discount, and **only moves to the next step if it's expected to be worth it**.

An uplift model decides who gets targeted. An expected-value model decides what they see and when.

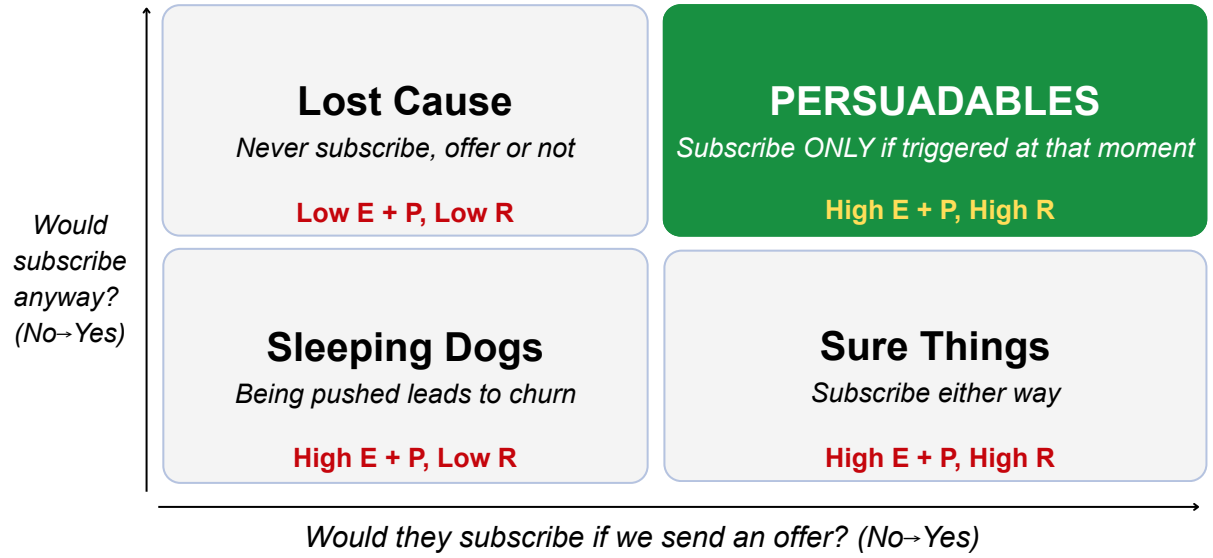
What makes this AI-driven rather than a fixed decision tree:

This system is not a fixed rules engine	What this system actually does
"If skip count > 10, show Rung 1" — a hand-written threshold	Stage 1 predicts each user's causal treatment effect, not just conversion likelihood
Same escalation path for every user, regardless of what actually motivates them	Stage 2 estimates expected value per rung, per person, before escalating
No estimate of whether the message itself changes behavior	Thresholds are learned and re-scored continuously against outcomes, not hand-set once
Recalibrating requires a person to rewrite the rule	

47% of free users say nothing would move them – the model finds who's left

Stage 1: Uplift scoring separates Lost Causes, Sleeping Dogs, and Sure Things from the Persuadables worth targeting

Free User Segmentation: An offer only creates value when it reaches a Persuadable



Lost Causes won't respond, Sure Things convert anyway, and Sleeping Dogs risk churning if pushed. Targeting only Persuadables means every offer has a real chance of changing the outcome, not just adding cost.

The algorithm tracks for to differentiate "Persuadables" – 3 Signals

E **EXTRACTION - *How much value are they taking?***

Key measures: Listening hours, Catalogue breadth, Saves and playlists built, Sessions per week

P **PRICE FIT - Which published plan fits them?**

Key measures: Market ARPU tier, Device tier, Plan eligibility (student, household), Payment method on file

R **READINESS - Are they at the wall right now?**

Key measures: Skip-cap hits, Blocked searches, Offline-save attempts, Ad-tolerance breaches, and their Acceleration over 7 days

Takes E/P/R scores and asks sharper "Will they convert because of the offer, versus anyway."

Decision Rule:

Low E + P, Low R	→	**Lost Cause**
High E + P, Low R	→	**Sleeping Dogs**
High E + P, High R	→	**Sure Things**
Moderate E + P, High R	→	***Persuadables***

Converting Users by Choosing the Right Rung, Timing, and Feature

Stage 2 decides what to do: choosing the rung, the timing, and the feature to show

Escalation Model

Key: The model only **escalates when** the **next step is worth trying**

Cost to Spotify Rises

The Action	Why This Matters
RUNG 0 | Do nothing No prompt at all applied to users that are not Persuadables.	**47% of Free users** say nothing would move them
RUNG 1 | Reveal the fit – proactive 93% of Free users are unaware of every feature Spotify has built since 2023. Direct, proactive suggestion make them more aware of how Premium addresses their problem.	**93% of Free users** unaware of any post-2023 Premium feature
RUNG 2 | Make friction visible – reactive "You hit the skip limit 14 times this month." Spotify already logs skip-cap hits, blocked searches, offline attempts and ad breaches.	**~100% of upgraders** hit a wall first
RUNG 3 | Let users experience it Single-use unlock – unlimited skips for this session, offline for this trip	**Leverages the "Endowment Effect"**
RUNG 4 | Route to the right plan Student, Duo, Family, Basic, annual. All publicly listed. Nothing bespoke.	**55% of current subs** already on discount plan.
RUNG 5 | Show published promo 3-month campaign anyone can claim — shown to who need it.	**33% + 53%** Free users say "definitely" or "maybe" to lower price

The end goal is net lifetime value: a user is only a win if what they generate over time outweighs what it cost to convert them, and if they stick around long enough to pay that back.

Before escalating, the model asks three questions:

1. **Did the last step already work?**
   If the user converted, stop – nothing more to do.
2. **Is the next step worth trying?**
   Escalate only if the ***expected gain beats the added cost***.
3. **Is escalating better than waiting?**
   Sometimes patience beats a bigger offer – the model weighs both.

Illustrative example, not an observed case

- A Persuadable sees **Rung 1**. No response.
- Model checks: is **Rung 2** worth it? Yes → Escalates.
- Still no response. Is **Rung 3** worth it? No. For this user, cost now outweighs gain, so it stops.

Not every user reaches Rung 5.

Where money enters, and where it does not

- **Rung 4:** A different plan Spotify already offers, same price, open to anyone.
- **Rung 5:** A discount that's already public. The model only decides who sees it and when, not what it costs.

Extensions to Core Solution

Conversion alone can't close every 2030 gap, two targets sit structurally outside what a conversion engine can move

Targets That Cannot be Addressed by Conversion Alone

1) Gross Margin

Converting a free user raises revenue, but the royalty share Spotify pays out per stream doesn't change. Gross margin is set by what's paid out.

→ *A shift in **revenue mix toward higher-margin verticals** is needed to optimize this number.*

2) Total MAUs

System 1 only converts users already on the platform. It does nothing for the top of the funnel: users who never joined, or joined and left.

→ *Reaching 1B MAU requires **acquiring and retaining** the free base itself.*

Adding two more extensions, built on the same engine, to close the gap

Core Persuadables Engine & Escalator Ladder Model + Two Extensions

System 3: The extension
Grows and retains the free base itself

System 2: The extension
Shifts revenue mix toward higher-margin verticals

System 1: The Core Solution–Persuadables Engine and Escalator Ladder Model
Directly lift free-to-premium conversion

2030 Targets It Closes

1B MAU target

Gross margin (35–40%)

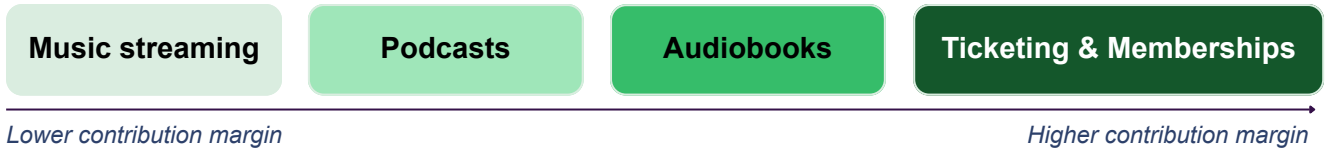
Revenue growth, contributes toward 1B MAU mix

System 2: Margin Mix Engine

System 2 helps redirecting uplift scoring toward revenue mix, not just conversion

System 2: Margin-Weighted Offer Allocation

Same engine, one extra term: the identical architecture from System 1, with content contribution margin added as a factor. Where a user has no strong preference between two options, the system leans toward the higher-margin one.



Key Rules

1. **Never applied to explicit intent**
   When the user has made an explicit choice themselves, like a search result, a direct play, or their own playlist, these are untouched.

2. **Lean-back surfaces only**
   Only applies when Spotify is already doing the algorithm and choosing on the user's behalf, like Home screen shelves or what plays next automatically

3. **Hard cap on rank displacement**
   The system can only choose between options it already thinks are roughly equally good for the user. Never promotes a worse fit for more margin.

4. **Publish exposure floors**
   Small nudges can add up in one direction over time, potentially favoring certain content over others. Spotify tracks and reports exposure, especially for independent artists, so any drift is visible in the data, not just assumed to be fine.

No rights holder can buy a rank. The purpose of this system is to **make smart, margin-aware suggestions**, *only when users has no clear preference*.

System 3: Growth & Retention Engine

Extending the same causal discipline to the two edges of the funnel Systems 1 and 2 don't reach

Why System 3?

- System 1 (Persuadables engine + Escalation Ladder) converts free users already engaged enough to show intent.
- System 2 shifts revenue mix once a user has no strong preference either way.

→ Both models have not considered new users and users who already left.

System 3 addresses that earlier and later edge of the funnel: **users just arriving**, and **users who've gone quiet**.

Accelerate MAU Growth + Decelerate Churn

The Gap	What AI can Do	What it improves
Brand-new users get weak recommendations *due to limited data; bad impression = early drop off*	**Cold-Start Personalization** *Technique: predictive modeling* Use early signals, region, device, language, first plays, personalize onboarding before enough history exists.	**Day-30 retention** Catch a new user's attention before they've decided Spotify "isn't for them."
Inactive users leave the platform *Inactive users don't announce they're leaving, they just quietly stop opening the app*	**Dormant Users Reactivation** *Technique: same causal uplift approach as System 1, but re-targeted* 1) Asking "***Would reaching out to this user cause them to come back***, or would they stay dormant regardless" 2) **Triggers a re-engagement message** surfacing what they used to listen to, timed to when someone's actually still recoverable, not fired at everyone who's gone quiet.	**Recovery & Visibility** Recovers users before permanently lost, and makes Spotify visible again at the moment it still has a chance to matter

Leverages predictive personalization and System 1's uplift engine to **make onboarding smarter** and **outreach better-timed**: **growing MAUs** by keeping new users past their first impression, and **reducing churn** by recovering dormant users before they're gone.

Implementation Roadmap: Sequencing the Three Systems to 2030

Build order follows dependency, risk, and value. Reuse the causal engine, prove conversion first, then extend to margin and reach

Time

YEAR 1
Pilot & Scale the Conversion Core

System 1

Persuadables Engine + Escalation Ladder

Decision

1. Ship System 1 first.
2. Launch on randomized holdout
3. Eventually scale to full free base

Why Implement

SPOTIFY'S GOAL ALIGNMENT

Free to Premium Conversion
Revenue Growth

- Lowest-cost path to revenue by reusing the uplift infrastructure Spotify already runs in production (2023)
- Builds the causal-measurement backbone that Systems 2 and 3 reuse.

YEAR 2
Add the margin layer

System 2

Margin Mix Engine

Decision

1. Add-on one margin term
2. Apply, with caution, on lean-back surfaces only

Why Implement

SPOTIFY'S GOAL ALIGNMENT

Gross margin 35-40%

- Identical architecture to System 1, so marginal build cost is near zero.
- Since royalty share per stream is fixed, only change in revenue mix can move gross margin

YEAR 3 until 2030
Extend to the funnel edges

System 3

Growth & Retention Engine

Decision

Retarget the same causal uplift to cold-start onboarding and dormant-user reactivation.

Why Implement

SPOTIFY'S GOAL ALIGNMENT

1B MAU
Lower churn rate

- Systems 1 and 2 only target already-engaged users.
- Reaching 1B MAU requires growing and retaining the free base itself

Year 1 Plan: Achieving Conversion within 12-Months with System 1

Standing up System 1 (Persuadables Engine + Escalation Ladder)

Time

Q1 | WEEKS 1-13
Foundation & Data

Goal

Stand up the data and team to score every free user

KEY ACTIVITIES

- Assemble the E / P / R signal pipeline (extraction, price-fit, readiness from logs Spotify already keeps.
- Lock the objective: margin-adjusted LTV with churn as constraint
- Design the randomized holdout and measurement framework

EXIT GATE – NEXT QUARTER

Signals validated end-to-end; holdout design signed off.

Q2 | WEEKS 14-26
Build & backtest

Goal

Stand up the data and team to score every free user

KEY ACTIVITIES

- Train the uplift model - treatment effect, not propensity - on E/P/ R.
- Build the expected-value engine that picks rung and timing per
- Backtest against historical conversions; calibrate thresholds
- Codify guardrails: no explicit intent, published exposure floors.

EXIT GATE – NEXT QUARTER

Backtest beats today's blanket-prompt baseline.

Q3 | WEEKS 27-39
Pilot on a holdout

Goal

Prove incremental conversion live, at small scale

KEY ACTIVITIES

- Launch on a randomized holdout in 1-2 markets
- Run the escalation rungs live; log the response at each step
- Measure incremental conversion vs control; watch churn and pressure.
- Iterate thresholds weekly against observed outcomes

EXIT GATE – NEXT QUARTER

Significant conversion lift, churn neutral or better.

Q4 | WEEKS 40-52
Foundation & Data

Goal

Roll out where uplift is proven; make it part of broader system

KEY ACTIVITIES

- Expand to the full free base in validated segments and markets
- Re-score continuously against outcomes — no hand-set thresholds
- Ship monitoring dashboards and exposure reporting
- Scope System 2 (margin term) for Year 2

EXIT GATE – NEXT QUARTER

Sustained lift at scale; handoff to a standing team.

Business Impact

Extending the same causal discipline to the two edges of the funnel Systems 1 and 2 don't reach

Business Impacts

Business and Financial Impact Summary *(Base case)*

Metric	Value
Five-year incremental revenue	€1,148.6M
NPV	€253.2M
ROI	289.6%
Revenue CAGR	14.5%
Total MAU	1.001B
Gross margin	35.3%
Operating margin	20.1%

Achieved 2030 Goals

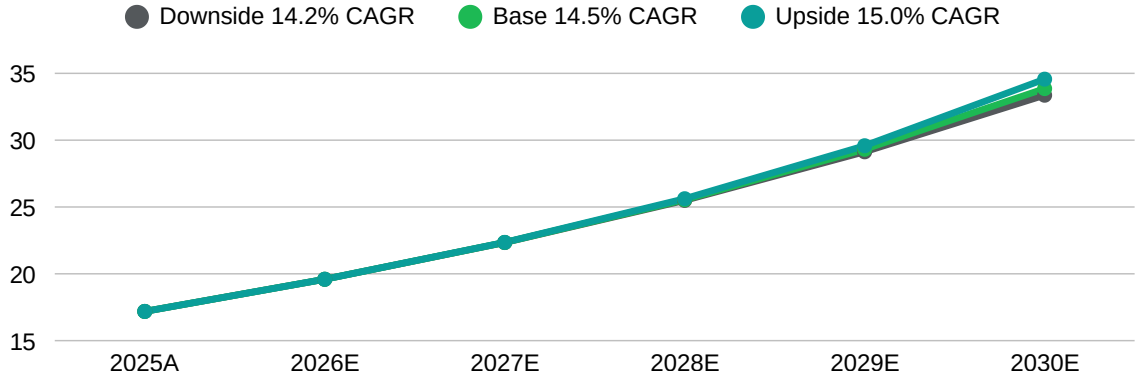
Key Assumptions

- **Cost assumption** | **Five year project cost: €148M Weak | €138M Base | €132M Strong**
  Includes direct system costs and one shared AI platform.
- **Weak Scenerio** | **Conversion +1 point | 2030 add on purchases +1 point | 30 day retention +2 points**
  Reactivation +2.5 points | Direct costs 15% above budget
- **Base Scenerio** | **Conversion +2 points | 2030 add on purchases +2 points | 30 day retention +4 points**
  Reactivation +5 points | Costs remain on budget
- **Strong Scenerio** | **Conversion +3 points | 2030 add on purchases +3 points | 30 day retention +6 points**
  Reactivation +7.5 points | Direct costs 10% below budget

Financials Breakdown

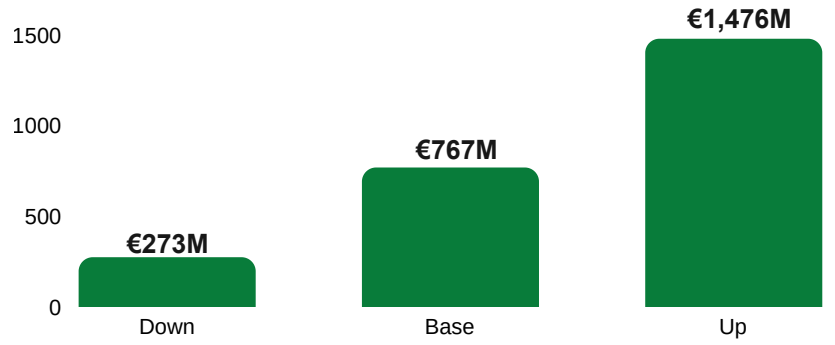
Revenue compounds at 14.2-15.0% CAGR across all three scenarios

Total Revenue Trajectory, in € billion



The solution is projected to add €767M in annual revenue by 2030, under the Base case

Incremental Revenue from Solution, in € million



WEAK SCENARIO
992M MAUs | 35.1% GM | 19.6% OM
STOP / REDESIGN — misses MAU and OM gates
Cumulative payback: 2030

EXPECTED SCENARIO
1.001B MAUs | 35.3% GM | 20.1% OM
STAGE-GATED SCALE — clears key thresholds
Cumulative payback: 2028

STRONG SCENARIO
1.010B MAUs | 35.6% GM | 20.7% OM
SCALE AFTER GATES — strongest value creation
Cumulative payback: 2028

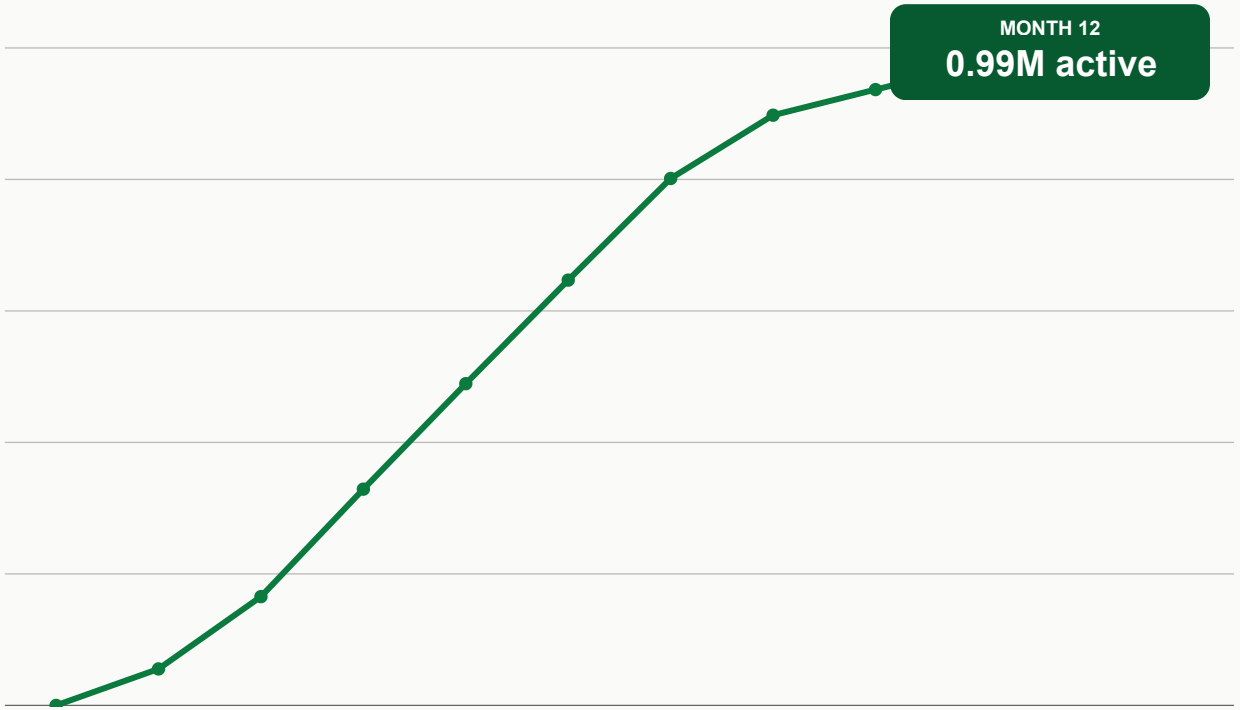
Business Impact

Our core solution (System 1) adds nearly 1.0M active Premium subscribers in 12 months

Active New Premium Subscribers

12-MONTH SYSTEM 1 PILOT

Millions of subscribers active at each month end



Month-12 Impact

WITH SYSTEM 1

+0.99M

additional active Premium subscribers

How The Model Get us to our Goal

55.5M users reached

× **2 more upgrades per 100 users**

= **1.11M Premium upgrades**

After Retention

Expected 0.99M remain active at Month 12

The conversion improvement stays constant; the Premium base grows as the pilot reaches more users.

Business Impact

The three systems help Spotify surpass 1 billion monthly active users and increase gross margin from 35.00% to 35.26%

Core solution helps conversion. System 3 is the user-growth engine; System 2 produces 74% of incremental gross benefit

2030 MAU Bridge

CORE SOLUTION	+	SYSTEM 3	TOTAL MAUS
983M		+17.6M	1.001B

2030 Gross Margin

TOTAL MARGIN
35.26%

***Achieved* Total MAUs**

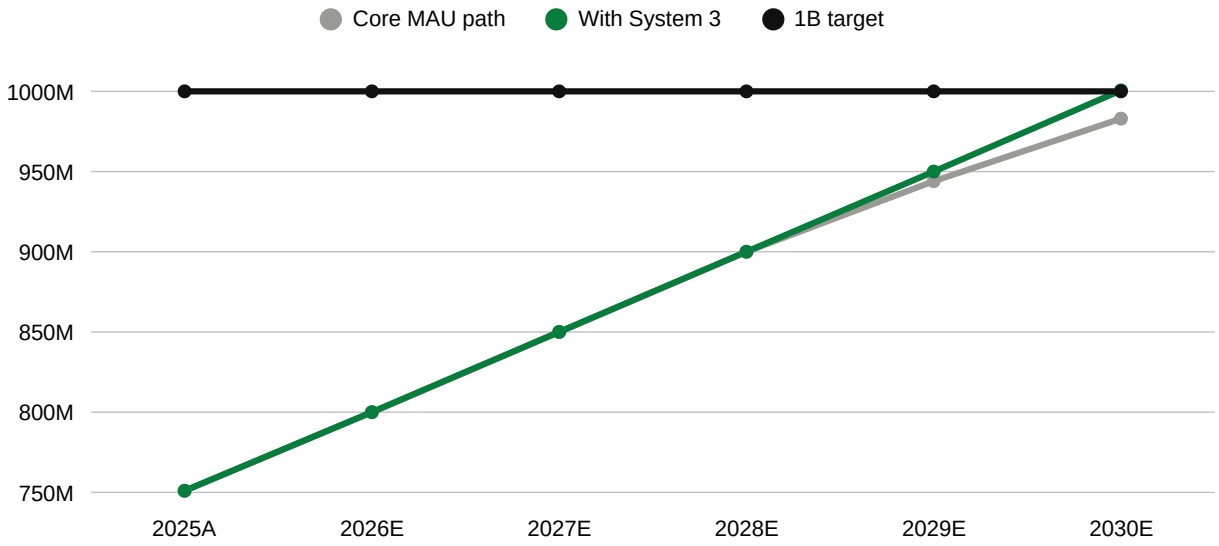
1 Billion by 2030

***Achieved* Gross Margin**

35%-40% Spotify's target range

System 3 closes the user gap while System 2 carries the margin economics

System 3 alone closes the gap between the core path (983M) and the 1B target



System 2 contributes €264.8M of the €357.3M total gross benefit

2030 incremental gross benefit by system (€M)



System 2 supplies 74% of gross benefit, making it the single biggest driver of margin improvement across all three systems.

Together, the three systems achieve both 2030 targets in the same year: System 3 delivers the user growth Spotify needs to cross 1 billion MAUs, while Systems 1 and 2 convert and monetize that larger base, reaching 35%+ gross margin.

Measurement Plan

Each system must beat its control group before Spotify invests further

Main Measure of Success

Positive additional profit

Base case: €400M net benefit after project costs; €253M NPV.

How it is Measured

Run a controlled pilot and compare results with similar users who do not receive the system. Scale only when the KPI beats the control group and the listening experience stays stable.

Metric	Control group / no system	Base-case KPI	Limit before pausing	Owner
System 1: Conversion and retention	No improvement vs. control	**+2 points; 80% remain after one year**	Improvement not proven or cancellations rise	Growth + Data Science
System 1: Promotional return	Current return on promotions	**€1.57 returned per €1 spent**	€1.00 or less for two quarters	Growth + Finance
System 2: Add-on adoption	No improvement vs. control	**+1 point in pilot; +2 by 2030**	Added gross profit turns negative	Product + Finance
System 2: Add-on economics	Current offer and price mix	**€75.60 revenue; 52.8% gross margin**	Satisfaction or artist exposure worsens	Finance + Pricing
System 3: Retention and return	No improvement vs. control	**+4 points at Day 30; +5 reactivated**	Improvement not proven vs. control	Markets + Growth
System 3: Cost per added user	No system-added users	**€1.68 per added active user**	More than €2.00 per user	Growth + Finance

Decision Rules – Set Before the Pilot

EXPAND *Every Base-case KPI holds for two quarters, added profit is positive, and user experience remains stable.*

PAUSE Any KPI misses its limit, cost rises above the cap, or listening, satisfaction, or artist exposure worsens.

Source: Spotify Three-System AI Financial Model, Base scenario. Model KPIs are assumptions that require controlled-pilot validation. “Points” means percentage points.

Risks and Mitigations

Every proposed system includes measurable guardrails to protect long-term ecosystem health.

Anticipated risks

	Description of Risk Factor	Evaluation of Severity	Risk Level	Mitigation
Discount as anchor, not incentive	• Personalized discounts may **anchor users to lower prices**. • When it ends, the user isn't comparing "no Premium vs. Premium" anymore, instead they're comparing **"Premium at X vs. Premium at full price"**.	• Full price now feels like **a loss**, and users even **value Spotify less** than before.	Medium-High	• Frame offers around Premium benefits rather than discounts. • When discounts are used, phase them out gradually. **Reinforce value of Premium** before the discount expires.
Free-tier erosion from over-targeting	• System 1 may **over-optimize for conversion** by increasing Premium prompts. • Overall MAUs can still appear healthy while individual Free users are **repeatedly over-targeted**, delaying detection of declining Free-tier engagement.	• **Threatens long-term user retention** and the **1B MAUs goal.**	High / Medium-High	• Limit how often the AI can show Premium offers (**frequency caps**) and pause promotions for users who repeatedly ignore or dismiss previous offers (**offer-fatigue score**).
Sleeping-dogs misclassification	• System 1 **incorrectly classifies** a Sleeping Dog as a Persuadable. • Model actively **targets someone it should leave alone.**	• **Reduced Free-tier engagement**, higher **churn risk**, and fewer future conversion opportunities.	High / Medium-High	• Continuously retrain the uplift model using **randomized holdout experiments**. • Use **confidence**, not just point estimate, in the policy.
Recommendation quality declines from margin optimization	• Margin-aware reranking gradually **favors lower-cost content** over the most relevant content, **reducing user satisfaction**. (System 2)	• **Reduced user engagement**, **increased churn**, and **weaker** long-term Premium retention.	Medium	• Keep **satisfaction-equivalence thresholds fixed.** • Gate C continuously **monitors ecosystem-health metrics** and **roll back System 2b** if predefined thresholds are breached.

Risks and Mitigations: Data Ethics

Every mitigation here uses data and infrastructure Spotify already has, no new collection, no new pricing mechanism, no new risk surface

Responsible Use of User Data and Behavioral Signals

Human-in-the-loop governance

Pricing and personalization decisions run through Spotify's existing revenue governance process, not a separate, hidden one.

Transparancy by default

Users can see and edit what the model infers about them - extending the existing Taste Profile pattern rather than inventing one.

Frequency caps and a fatigue score

The Free experience is never degraded by upsell pressure. Contact budget is enforced globally, across every team.

No person-to-person price differences

Every price shown comes from a single public price list, the same plan-matching design used throughout this solution. No one pays a different price than another user on the same plan.

No new data collection

Everything used is behavioural exhaust already collected under existing consent. We add an objective, not a microphone.

No dark patterns - stated explicitly

No countdown pressure, no hidden cancellation, no confirm-shaming. Ruled out explicitly, not just avoided by default.

Every guardrail here uses consent, infrastructure, and governance Spotify already has, this is disciplined use of existing capability, not a new risk surface

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Appendix A1 – Integrated Financial Model: Revenue, Profitability, and MAU Bridge

Spotify

Corporate Revenue, Margin and MAU Bridge							
Metric	Unit	2025A	2026E	2027E	2028E	2029E	2030E
Core revenue	€m	€17,186.0	€19,592.0	€22,334.9	€25,461.8	€29,026.5	€33,090.2
System 1 revenue	€m	-	-	€9.6	€55.5	€120.3	€204.2
System 2 revenue	€m	-	-	-	€26.5	€151.4	€476.6
System 3 revenue	€m	-	-	-	-	€18.0	€86.4
Total revenue incl. solution	**€m**	**€17,186.0**	**€19,592.0**	**€22,344.6**	**€25,543.8**	**€29,316.2**	**€33,857.4**
15% target revenue path	**€m**	**€17,186.0**	**€19,763.9**	**€22,728.5**	**€26,137.8**	**€30,058.4**	**€34,567.2**
Revenue gap / (surplus) to target	**€m**	**-**	**€171.9**	**€383.9**	**€593.9**	**€742.2**	**€709.8**
Share of 14%-to-15% gap closed	%	**0.0%**	**0.0%**	**2.4%**	**12.1%**	**28.1%**	**51.9%**
Revenue CAGR incl. solution	%	**0.0%**	**14.0%**	**14.0%**	**14.1%**	**14.3%**	**14.5%**
Core gross profit	€m	€5,496.1	€6,465.4	€7,482.2	€8,657.0	€10,043.2	€11,581.6
System 1 net gross benefit	€m	-	-	€3.3	€19.0	€41.5	€70.9
System 2 gross benefit	€m	-	-	-	€15.5	€85.7	€264.8
System 3 gross profit	€m	-	-	-	-	€4.5	€21.6
Total gross profit	**€m**	**€5,496.1**	**€6,465.4**	**€7,485.5**	**€8,691.5**	**€10,174.9**	**€11,938.8**
Gross margin	%	**32.0%**	**33.0%**	**33.5%**	**34.0%**	**34.7%**	**35.3%**
Core operating income	€m	€2,199.8	€2,938.8	€3,685.3	€4,455.8	€5,398.9	€6,485.7
System 1 contribution	€m	-	-	(€3.4)	€11.0	€32.4	€59.7
System 2 contribution	€m	-	-	-	€8.4	€75.4	€250.4
System 3 contribution	€m	-	-	-	-	(€8.2)	€0.1
Shared platform cost	€m	-	€4.0	€4.0	€5.0	€6.0	€7.0
Total operating income	**€m**	**€2,199.8**	**€2,934.8**	**€3,677.8**	**€4,470.2**	**€5,492.5**	**€6,788.9**
Operating margin	%	**12.8%**	**15.0%**	**16.5%**	**17.5%**	**18.7%**	**20.1%**
Core MAUs before System 3	millions	751.0	800.0	850.0	900.0	944.0	983.0
System 3 active incremental MAUs	millions	-	-	-	-	6.0	17.6
Total MAUs incl. System 3	millions	751.0	800.0	850.0	900.0	950.0	1,000.6
Ad MAUs before System 1	millions	476.0	510.0	535.0	575.0	620.0	665.0
System 3 incremental Free MAUs	millions	-	-	-	-	5.4	15.8
System 1 active converts	millions	-	-	0.5	1.6	3.0	4.8
Free / Ad MAUs after initiatives	millions	476.0	510.0	534.5	573.4	622.4	676.1
Free share of total MAUs	%	63.4%	63.8%	62.9%	63.7%	65.5%	67.6%
Revenue / MAU	€ / MAU	€22.88	€24.49	€26.29	€28.38	€30.86	€33.84
Revenue / MAU multiple vs Q1 2026	x	1.01x	1.08x	1.16x	1.26x	1.37x	1.50x
Total direct system cost	€m	-	-	€7.7	€17.4	€35.4	€51.5
Total project cost incl. shared platform	**€m**	**-**	**€4.0**	**€11.7**	**€22.4**	**€41.4**	**€58.5**
Net initiative contribution after shared platform	**€m**	**-**	**(€4.0)**	**(€7.4)**	**€14.4**	**€93.5**	**€303.2**
Cumulative initiative contribution	**€m**	**-**	**(€4.0)**	**(€11.4)**	**€3.0**	**€96.5**	**€399.7**
PV of initiative contribution	**€m**	**-**	**(€3.6)**	**(€6.1)**	**€10.8**	**€63.9**	**€188.2**

Appendix A2: Scenario Assumptions

Scenario Assumptions						
Driver	Unit	Downside	Base	Upside	Active	Why / validation
System 1 treatment-control lift	percentage points	1.00%	2.00%	3.00%	2.00%	Causal scale threshold; NPEP calibrates trend but pilot validates lift.
System 1 annual retention	%	70.0%	80.0%	85.0%	80.0%	Retained incremental subscribers, not raw conversions.
System 1 target cohort multiplier	x base rollout	0.75x	1.00x	1.20x	1.00x	Controls size of E/P/R-selected persuadable pool.
System 1 ARPU realization	% of reported ARPU	85.0%	90.0%	95.0%	90.0%	Captures market mix, plan mix and offer discounts.
System 1 promo-waste recovery	%	30.0%	50.0%	70.0%	50.0%	Savings from no-offer and avoiding sure-things/lost-causes.
System 2 attach uplift multiplier	x base rollout	0.50x	1.00x	1.50x	1.00x	Scales incremental add-on purchase/attach lift versus control.
System 2 revenue-per-attach factor	x offer matrix	0.75x	1.00x	1.25x	1.00x	Tests product and price mix around the offer portfolio.
System 2 GM adjustment	percentage points	-5.00%	0.00%	5.00%	0.00%	Sensitivity around weighted offer contribution margin.
System 2 eligible-base margin uplift	percentage points	0.50%	1.00%	1.50%	1.00%	Reranking mix shift inside the narrow eligible revenue base only.
System 3 cold-start D30 retention uplift	percentage points	2.00%	4.00%	6.00%	4.00%	Must be isolated against holdout in under-penetrated markets.
System 3 dormant reactivation uplift	percentage points	2.50%	5.00%	7.50%	5.00%	Incremental reactivation versus control.
System 3 annual retention	%	70.0%	80.0%	85.0%	80.0%	Retention of the incremental MAU cohorts.
System 3 annual revenue / active MAU	€/MAU/year	€4.50	€6.00	€7.50	€6.00	Blended low-ARPU monetization; does not assume full Spotify average.
Direct-cost multiplier	x	1.15x	1.00x	0.90x	1.00x	Applies to direct system costs; protects downside feasibility.
Discount rate	%	10.0%	10.0%	10.0%	10.0%	Project hurdle rate for NPV.

Appendix A3: Key Assumption Stress Tests Across the Three Systems

System 1 - 12M Net Contribution (€m): Reach vs Causal Lift					
Reach \ Lift	1.00%	2.00%	3.00%	4.00%	5.00%
30.0%	(€4.5)	(€2.6)	(€0.6)	€1.3	€3.3
50.0%	(€3.8)	(€0.5)	€2.7	€6.0	€9.2
70.0%	(€3.0)	€1.5	€6.1	€10.6	€15.1

System 2 - 2030 Net Contribution (€m): Attach Lift vs Revenue / Attach					
Attach Lift \ Revenue	€50.00	€70.00	€90.00	€110.00	€130.00
1.00%	€81.5	€114.6	€147.7	€180.7	€213.8
2.00%	€164.2	€230.3	€296.4	€362.5	€428.6
3.00%	€246.8	€346.0	€445.1	€544.3	€643.5

System 3 - 2030 Cost / New Net MAU (€): Cold-Start Lift vs Cost / Reached User					
D30 Lift \ Cost	€0.02	€0.03	€0.04	€0.05	€0.06
2.00%	€2.30	€2.56	€2.82	€3.08	€3.35
4.00%	€1.51	€1.68	€1.85	€2.02	€2.20
6.00%	€1.12	€1.25	€1.38	€1.51	€1.63

Appendix A4 – 2030 Goal Check: Targets Compared with Model Results

Goal	Target	Model Output	Status	Primary System
1B MAUs	1,000.0	1,000.6	PASS	System 3
Free above 50%	50.0%	67.6%	PASS	Unified platform
Mid-teens revenue growth	15.0%	14.5%	PARTIAL	Systems 1 + 2
Revenue / MAU	€34.60	€33.84	PARTIAL	Systems 1 + 2
Revenue / MAU multiple	1.50x	1.50x	PARTIAL	Systems 1 + 2
35-40% gross margin	35.0%	35.3%	SUPPORTS	System 2
20%+ operating margin	20.0%	20.1%	SUPPORTS	Systems 1 + 2
Preserve Free experience	2.0%	0.7%	PASS	System 1

Appendix A5 | Survey Link

Link: https://docs.google.com/spreadsheets/d/1JPAZvbO4fSLx2T7nxt_MdG5hpgoafDqQDLv5A5YPumk/edit?usp=sharing

Timestamp	How old are	Where do you live?	On average, how many hours per day do you
7/15/2026 10:11:14	18–24	Asia	2-4 hours
7/15/2026 10:14:22	18–24	Asia	Over 4 hours
7/15/2026 10:16:22	18–24	North America	1-2 hours
7/15/2026 10:17:04	25–34	Europe	1-2 hours
7/15/2026 10:33:12	18–24	North America	Over 4 hours
7/15/2026 10:33:32	18–24	Asia	2-4 hours
7/15/2026 10:34:17	18–24	Asia	Over 4 hours
7/15/2026 10:35:29	25–34	Asia	Under 1 hour
7/15/2026 10:44:33	18–24	North America	Under 1 hour
7/15/2026 10:45:13	18–24	Asia	2-4 hours
7/15/2026 10:47:08	18–24	Asia	1-2 hours
7/15/2026 10:50:23	18–24	Asia	2-4 hours

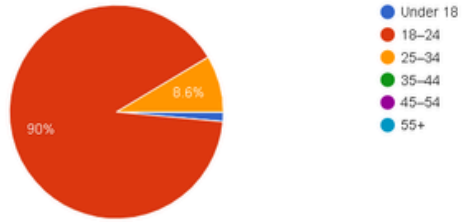
Form Responses 1 | Current subscribers | Sheet5 | Former subscribers | Free users | Never use

Appendix A6: Survey Data – Demographic Responses Overview

How old are you?

Copy chart

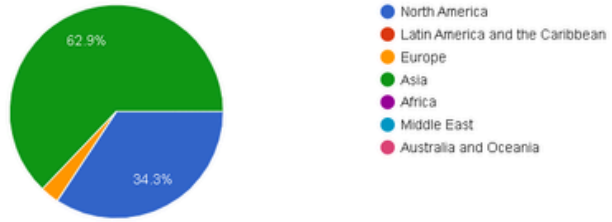
70 responses



Where do you live?

Copy chart

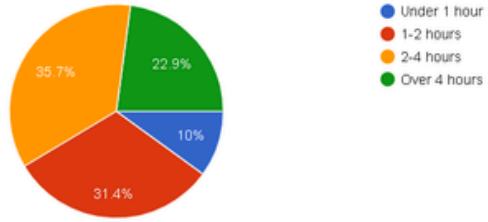
70 responses



On average, how many hours per day do you listen to music or audio?

Copy chart

70 responses



Which audio streaming service do you use most often?

Copy chart

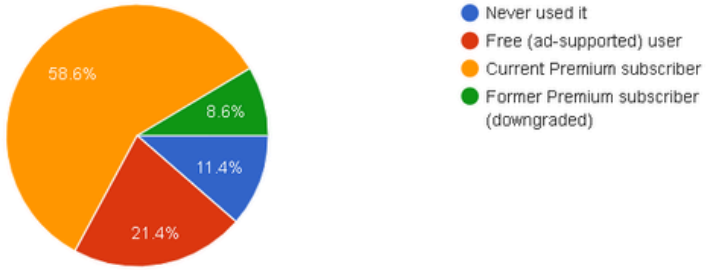
70 responses



If you are using Spotify, which best describes your current Spotify status?

Copy chart

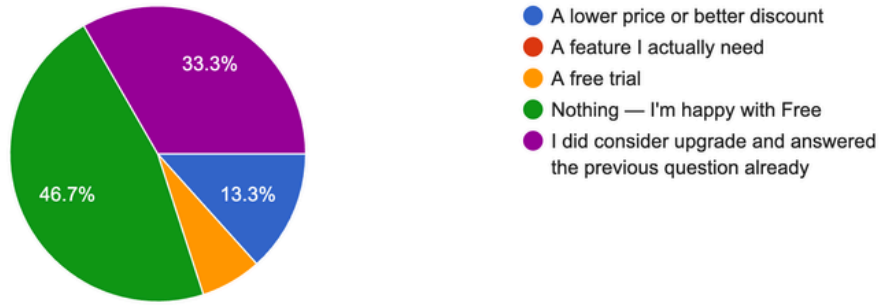
70 responses



Appendix A6: Survey Data – Behavioral Responses Overview

If No, what would it take for you to consider upgrade?

15 responses



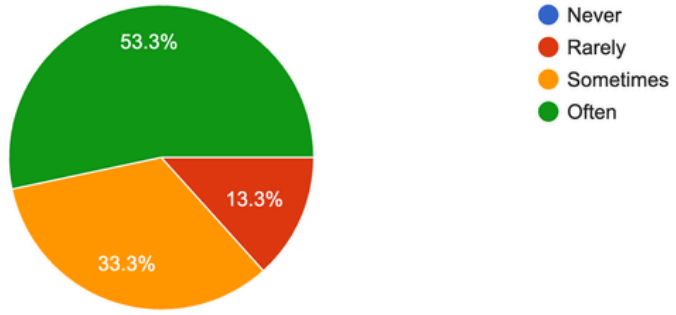
(If Yes) What finally made you decide to upgrade?

41 responses



How often do ads or skip limits interrupt your listening in a way that actually bothers you?

15 responses



(If Yes) In the weeks right before you upgraded, was your listening...

41 responses

